

JOINS

```
CREATE TABLE DEPT(
DEPARTMENTID INT PRIMARY KEY,
DEPARTMENTNAME VARCHAR(100) NOT NULL UNIQUE,
DEPARTMENTCODE VARCHAR(50) NOT NULL UNIQUE,
LOCATION VARCHAR(50) NOT NULL)

INSERT INTO DEPT VALUES (1, 'ADMIN', 'ADM', 'A-BLOCK')
INSERT INTO DEPT VALUES (2, 'COMPUTER', 'CE', 'C-BLOCK')
INSERT INTO DEPT VALUES (3, 'CIVIL', 'CI', 'G-BLOCK')
INSERT INTO DEPT VALUES (4, 'ELECTRICAL', 'EE', 'E-BLOCK')
INSERT INTO DEPT VALUES (5, 'MECHANICAL', 'ME', 'B-BLOCK')

DROP TABLE PERSON
CREATE TABLE PERSON(
PERSONID INT PRIMARY KEY,
PERSONNAME VARCHAR(100) NOT NULL,
DEPARTMENTID INT NULL FOREIGN KEY REFERENCES DEPT(DEPARTMENTID),
SALARY DECIMAL(8,2) NOT NULL,
JOININGDATE DATETIME NOT NULL,
CITY VARCHAR(100) NOT NULL)

INSERT INTO PERSON VALUES (101, 'RAHUL TRIPATHI', 2, 56000, '01-01-2000', 'RAJKOT')
INSERT INTO PERSON VALUES (102, 'HARDIK PANDYA', 3, 18000, '25-SEP-2001', 'AHMEDABAD')
INSERT INTO PERSON VALUES (103, 'BHAVIN KANANI', 4, 25000, '14-MAY-2000', 'BARODA')
INSERT INTO PERSON VALUES (104, 'BHOOMI VAISHNAV', 1, 39000, '08-FEB-2005', 'RAJKOT')
INSERT INTO PERSON VALUES (105, 'ROHIT TOPIYA', 2, 17000, '23-JUL-2001', 'JAMNAGAR')
INSERT INTO PERSON VALUES (106, 'PRIYA MENPARA', NULL, 9000, '18-OCT-2000', 'AHMEDABAD')
INSERT INTO PERSON VALUES (107, 'NEHA SHARMA', 2, 34000, '25-DEC-2002', 'RAJKOT')
INSERT INTO PERSON VALUES (108, 'NAYAN GOSWAMI', 3, 25000, '01-JUL-2001', 'RAJKOT')
INSERT INTO PERSON VALUES (109, 'MEHUL BHUNDIYA', 4, 13500, '09-JAN-2005', 'BARODA')
INSERT INTO PERSON VALUES (110, 'MOHIT MARU', 5, 14000, '25-MAY-2000', 'JAMNAGAR')

SELECT * FROM DEPT
SELECT * FROM PERSON
----- SIMPLE JOINS-----
--1 LIST ALL PEOPLE ALONG WITH THEIR DEPARTMENT NAMES.
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
full outer JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;

--2 SHOW EACH PERSON'S NAME, SALARY, AND DEPARTMENT.
SELECT
    PERSON.PERSONNAME,
    PERSON.SALARY,
    DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;
```

--3 RETRIEVE EACH PERSON'S NAME, CITY, AND THE BLOCK LOCATION OF THEIR DEPARTMENT.

```
SELECT
    PERSON.PERSONNAME,
    PERSON.CITY,
    DEPT.LOCATION
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID;
```

--4 DISPLAY PERSONID, NAME, JOINING DATE AND DEPTNAME.

```
SELECT
    PERSON.PERSONID,
    PERSON.PERSONNAME,
    PERSON.JOININGDATE,
    DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

-----SIMPLE JOIN + WHERE-----

--5 DISPLAY THE NAME OF PEOPLE WORK IN THE COMPUTER DEPARTMENT

```
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTNAME = 'COMPUTER';
```

--6 DISPLAY PEOPLE LIVE IN RAJKOT, AND WHAT'S THEIR DEPARTMENT?

```
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME,
    PERSON.CITY
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT';
```

--7 DISPLAY PERSON NAME WHO IS SITTING IN 'C-BLOCK'

```
SELECT PERSON.PERSONNAME,DEPT.LOCATION
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE DEPT.LOCATION='C-BLOCK'
```

----- JOIN + WHERE + MULTIPLE CONDITIONS -----

--8 WHO EARNS MORE THAN 15,000 AND LIVES IN AHMEDABAD

--DISPLAY THE DETAIL WITH THEIR DEPARTMENT NAME?

SELECT

PERSON.PERSONNAME,
PERSON.SALARY,
PERSON.CITY,
dept.DepartmentName

FROM PERSON

JOIN DEPT

ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE PERSON.SALARY > 15000 **AND** PERSON.CITY = 'AHMEDABAD';

--9 WHICH PERSON BELONG TO DEPARTMENTS WITH CODES 'CE' OR 'EE'?

SELECT

PERSON.PERSONNAME,
DEPT.DEPARTMENTCODE

FROM PERSON

JOIN DEPT

ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE DEPT.DEPARTMENTCODE **IN** ('CE', 'EE');

--10 WHO EARNS A SALARY NOT EQUAL TO 17,000

--AND DOES NOT LIVE IN BARODA

--DISPLAY WITH DEPTNAME.

SELECT

PERSON.PERSONNAME,
PERSON.SALARY,
DEPT.DEPARTMENTNAME

FROM PERSON

JOIN DEPT

ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE PERSON.SALARY != 17000 **AND** PERSON.CITY <> 'BARODA';

--11 FIND PEOPLE WITH DEPT NAME WHOSE SALARY IS BETWEEN 20,000 AND 40,000

SELECT

PERSON.PERSONNAME,
PERSON.SALARY

FROM PERSON

JOIN DEPT

ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE PERSON.SALARY **BETWEEN** 20000 **AND** 40000;

--12 WHO WORKS IN ANY DEPARTMENT WHOSE NAME STARTS WITH 'C'

--AND EARNS MORE THAN 10,000?

SELECT

PERSON.PERSONNAME,
DEPT.DEPARTMENTNAME,
PERSON.SALARY

FROM PERSON

JOIN DEPT

ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE DEPT.DEPARTMENTNAME **LIKE** 'C%' **AND** PERSON.SALARY > 10000;

--13 DISPLAY PERSON WHO JOINED THE CIVIL DEPARTMENT AFTER 1-AUG-2001

SELECT PERSON.PERSONNAME, DEPT.DEPARTMENTNAME, PERSON.JOININGDATE

```
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE DEPT.DEPARTMENTNAME='CIVIL'
AND PERSON.JOININGDATE>'1-AUG-2001'

-----JOIN + GROUP BY-----
--14 WHAT IS THE AVERAGE SALARY OF EACH DEPARTMENT

SELECT
    DEPT.DEPARTMENTNAME,
    AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;

--15 COUNT HOW MANY PEOPLE ARE WORKING IN EACH DEPARTMENT.
SELECT
    DEPT.DEPARTMENTNAME,
    COUNT(PERSON.PERSONID) AS NUMPEOPLE
FROM DEPT
JOIN PERSON
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;

--16 DISPLAY TOTAL SALARIES CITY WISE.
SELECT
    PERSON.CITY,
    SUM(PERSON.SALARY) AS TOTAL_SAL
FROM PERSON
GROUP BY PERSON.CITY

-- JOIN + WHERE + GROUP BY
--17 WHAT'S THE MINIMUM SALARY PER DEPARTMENT FOR PEOPLE LIVE IN RAJKOT?
SELECT
    DEPT.DEPARTMENTNAME,
    MIN(PERSON.SALARY) AS MIN_SAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT'
GROUP BY DEPT.DEPARTMENTNAME;

--18 COUNT PEOPLE IN IN EACH DEPARTMENT LIVES IN AHMEDABAD
SELECT
    DEPT.DEPARTMENTNAME,
    COUNT(*) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'AHMEDABAD'
GROUP BY DEPT.DEPARTMENTNAME;
```

--19 SUM OF SALARIES FOR PERSON WHO JOINED BEFORE 2001, IN EACH DEPARTMENT.

```
SELECT
    SUM(PERSON.SALARY) AS TOTAL,
    DEPT.DEPARTMENTNAME
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
WHERE PERSON.JOININGDATE < '2001-01-01'
GROUP BY DEPT.DEPARTMENTNAME
```

-- JOIN + GROUP BY + HAVING---

--20 WHICH DEPARTMENTS HAVE AN AVERAGE SALARY HIGHER THAN 30,000

```
SELECT
    DEPT.DEPARTMENTNAME,
    AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING AVG(PERSON.SALARY) > 30000;
```

--21 DISPLAY DEPARTMENTS WHERE MORE THAN 2 EMPLOYEES ARE WORKING.

```
SELECT
    DEPT.DEPARTMENTNAME,
    COUNT(*) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING COUNT(*) > 2;
```

--22 FIND ALL DEPARTMENTS WHOSE TOTAL SALARY IS EXCEEDING 100000

```
SELECT DEPT.DEPARTMENTNAME, SUM(PERSON.SALARY)
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME
HAVING SUM(PERSON.SALARY)>100000
```

--JOIN + WHERE + GROUP BY + HAVING

--23) AMONG PEOPLE IN RAJKOT, WHICH DEPARTMENTS HAVE AVG SALARY
--LESS THEN 30,000?

```
SELECT
    DEPT.DEPARTMENTNAME,
    AVG(PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'RAJKOT'
GROUP BY DEPT.DEPARTMENTNAME
HAVING AVG(PERSON.SALARY)<30000;
```

--24 IN AHMEDABAD, WHICH DEPARTMENTS HAVE ONLY ONE EMPLOYEE?

```
SELECT
    DEPT.DEPARTMENTNAME,
    COUNT (PERSON.PERSONID) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
WHERE PERSON.CITY = 'AHMEDABAD'
GROUP BY DEPT.DEPARTMENTNAME
HAVING COUNT (PERSON.PERSONID) = 1;
```

---25 GROUPING ON MULTIPLE COLUMN-----

--HOW MANY EMPLOYEES ARE THERE IN EACH CITY AND EACH DEPARTMENT?

```
SELECT
    PERSON.CITY,
    DEPT.DEPARTMENTNAME,
    COUNT (PERSON.PERSONID) AS NUMEMP
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY PERSON.CITY, DEPT.DEPARTMENTNAME;
```

--26 JOIN WITH 2 AGGREGATE FUNCTIONS

--SHOW THE PERSON COUNT AND MAXIMUM SALARY PER DEPARTMENT.

```
SELECT DEPT.DEPARTMENTNAME,
    COUNT (PERSON.PERSONID) AS NUMEMP,
    MAX (PERSON.SALARY) AS MAXSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;
```

--27 SHOW THE TOTAL AND AVERAGE SALARY PER DEPARTMENT.

```
SELECT DEPT.DEPARTMENTNAME,
    SUM (PERSON.SALARY) AS TOTALSAL,
    AVG (PERSON.SALARY) AS AVGSAL
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID
GROUP BY DEPT.DEPARTMENTNAME;
```

--28 CONCATENATE STRING WITH JOIN

--PRODUCE OUTPUT LIKE: <PERSONNAME> LIVES IN <CITY>

--AND WORKS IN <DEPARTMENTNAME> DEPARTMENT AND EARNS <SALARY> (IN SINGLE COLUMN)

```
SELECT
    PERSON.PERSONNAME + ' LIVES IN ' + PERSON.CITY + ' AND WORKS IN '
+ DEPT.DEPARTMENTNAME + ' DEPARTMENT AND EARNS ' +
    CAST (PERSON.SALARY AS VARCHAR)
FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

```
--29 UPDATE WITH JOIN
--INCREASE THE SALARY BY 10% FOR ALL EMPLOYEES WHOSE DEPARTMENT CODE 'CE'.
```

```
UPDATE PERSON
SET PERSON.SALARY = PERSON.SALARY+ (PERSON.SALARY * 0.1)

FROM PERSON
JOIN DEPT
ON PERSON.DEPARTMENTID = DEPT.DEPARTMENTID

WHERE DEPT.DEPARTMENTCODE = 'CE';
```

```
---JOIN WITH ALIAS-----
--30 DISPLAY PERSON NAME AND DEPARTMENT NAME
SELECT
    P.PERSONNAME,
    D.DEPARTMENTNAME
FROM PERSON P
JOIN DEPT D
ON P.DEPARTMENTID=D.DEPARTMENTID
```

```
-----TYPES OF JOIN-----
---31 INNER JOIN---
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
INNER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

```
---32 LEFT OUTER JOIN----
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
LEFT OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

```
---33 RIGHT OUTER JOIN----
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
RIGHT OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID
```

```
---34 FULL OUTER JOIN----
SELECT
    PERSON.PERSONNAME,
    DEPT.DEPARTMENTNAME
FROM PERSON
```

```
FULL OUTER JOIN DEPT
ON PERSON.DEPARTMENTID=DEPT.DEPARTMENTID

---35 CROSS JOIN---
SELECT *
FROM PERSON
CROSS JOIN
DEPT
--OR
SELECT *
FROM PERSON, DEPT

---SELF JOIN---
CREATE TABLE Employe(
EmpNo    VARCHAR(6)      PRIMARY KEY,
Name     VARCHAR(50)     NOT NULL,
MngrNo   VARCHAR(6)      NULL)

INSERT INTO Employe (EmpNo, Name, MngrNo) VALUES
('E00004', 'Milan',    NULL),
('E00005', 'Jay',      NULL),
('E00002', 'Rohan',    'E00005'),
('E00003', 'Priya',    'E00005'),
('E00001', 'Tarun',    'E00002'),
('E00006', 'Anjana',   'E00003');

SELECT * FROM Employe

--36 TO RETRIEVE THE LIST OF EMPLOYEES NAME AND THEIR MANAGERS NAME
SELECT
    E.NAME AS EMPLOYE,
    M.NAME AS MANAGER
FROM EMPLOYE E
INNER JOIN EMPLOYE M
ON E.MNGRNO=M.EMPNO

-----3 TABLES JOIN-----
-- AUTHORS TABLE
CREATE TABLE AUTHORS (
    AUTHORID INT PRIMARY KEY,
    AUTHORNAME VARCHAR(100),
    BIRTHDATE DATE
);
INSERT INTO AUTHORS (AUTHORID, AUTHORNAME, BIRTHDATE)
VALUES
(1, 'RABINDRANATH TAGORE', '1861-05-07'),
(2, 'CHETAN BHAGAT', '1974-04-22'),
(3, 'ARUNDHATI ROY', '1961-11-24');

-- CREATING THE PUBLISHERS TABLE
CREATE TABLE PUBLISHERS (
    PUBLISHERID INT PRIMARY KEY,
    PUBLISHERNAME VARCHAR(100),
```

```
COUNTRY VARCHAR(50)
);
INSERT INTO PUBLISHERS (PUBLISHERID, PUBLISHERNAME, COUNTRY)
VALUES
(1, 'RUPA PUBLICATIONS', 'INDIA'),
(2, 'PENGUIN RANDOM HOUSE INDIA', 'INDIA'),
(3, 'HARPERCOLLINS INDIA', 'INDIA');

-- CREATING THE BOOKS TABLE
CREATE TABLE BOOKS (
    BOOKID INT PRIMARY KEY,
    TITLE VARCHAR(200),
    PUBLISHEDYEAR DATETIME,
    AUTHORID INT REFERENCES AUTHORS(AUTHORID),
    PUBLISHERID INT REFERENCES PUBLISHERS(PUBLISHERID)
);
INSERT INTO BOOKS
VALUES
(1, 'GITANJALI', '1-JAN-1910', 1, 1),
(2, 'FIVE POINT SOMEONE', '1-JAN-2004', 2, 1),
(3, 'THE GOD OF SMALL THINGS', '1-JAN-1997', 3, 2);

--37 DISPLAY BOOK TITLE, AUTHERNAME, PUBLISHER NAME, AND PUBLISHED YEAR
SELECT
    BOOKS.TITLE,
    BOOKS.PUBLISHEDYEAR,
    AUTHORS.AUTHORNAME,
    PUBLISHERS.PUBLISHERNAME
FROM
    BOOKS
    JOIN AUTHORS
    ON BOOKS.AUTHORID = AUTHORS.AUTHORID

    JOIN PUBLISHERS
    ON BOOKS.PUBLISHERID = PUBLISHERS.PUBLISHERID;

--38 DISPLAY BOOK TITLE, AUTHOR NAME, PUBLISHED YEAR
--AND PUBLISHER'S COUNTRY
SELECT
    BOOKS.TITLE,
    AUTHORS.AUTHORNAME,
    PUBLISHERS.COUNTRY,
    BOOKS.PUBLISHEDYEAR

FROM
    BOOKS
    JOIN AUTHORS
    ON BOOKS.AUTHORID=AUTHORS.AUTHORID

    JOIN PUBLISHERS
    ON BOOKS.PUBLISHERID=PUBLISHERS.PUBLISHERID

--39 INCLUDE ALL BOOKS EVEN IF AUTHOR OR PUBLISHER IS MISSING
SELECT
    B.TITLE,
    A.AUTHORNAME,
    P.PUBLISHERNAME,
```

```
B.PUBLISHEDYEAR
FROM BOOKS B
LEFT JOIN AUTHORS A
  ON B.AUTHORID = A.AUTHORID
LEFT JOIN PUBLISHERS P
  ON B.PUBLISHERID = P.PUBLISHERID;

--40 ALL THE DETAILS OF BOOKS PUBLISHED AFTER 1990 IN DESC ORDER

SELECT
  B.TITLE,
  A.AUTHORNAME,
  A.BIRTHDATE,
  P.PUBLISHERNAME,
  P.COUNTRY,
  B.PUBLISHEDYEAR
FROM BOOKS B
INNER JOIN AUTHORS A
  ON B.AUTHORID = A.AUTHORID
INNER JOIN PUBLISHERS P
  ON B.PUBLISHERID = P.PUBLISHERID

WHERE B.PUBLISHEDYEAR > '1990-12-31'
ORDER BY B.PUBLISHEDYEAR DESC;

----NO OF RECORDS RETURNS BY DIFFERENT TYPES OF JOINS----
CREATE TABLE T1
(N01 INT)

INSERT INTO T1 VALUES (1), (1), (1), (2), (3), (3), (3)

SELECT * FROM T1
SELECT * FROM T2

CREATE TABLE T2
(N01 INT)

INSERT INTO T2 VALUES (1), (1), (2), (2), (4), (NULL)

--INNER JOIN
SELECT *
FROM T1
INNER JOIN T2
ON T1.N01=T2.N01
select * from t1
select * from t2

--LEFT OUTER JOIN
SELECT *
FROM T1
LEFT OUTER JOIN T2
ON T1.N01=T2.N01
```

```
---RIGHT OUTER JOIN
```

```
SELECT *  
FROM T1  
RIGHT OUTER JOIN T2  
ON T1.NO1=T2.NO1
```

```
---FULL OUTER JOIN--
```

```
--INNER JOIN
```

```
SELECT *  
FROM T1  
FULL OUTER JOIN T2  
ON T1.NO1=T2.NO1
```

```
insert INTO T1 VALUES (null)  
insert INTO T2 VALUES (null)
```